

Unit 1 – Exploring One-Variable Data
Notes 2 – Representing Quantitative Variables with Graphs

Graphing Quantitative Variables: Histograms

The table below shows the average income level for a household in the respective state.

State	Average Income	State	Average Income
Alabama	\$78,871.29	Montana	\$81,638.21
Alaska	\$100,086.45	Nebraska	\$93,878.49
Arizona	\$96,364.72	Nevada	\$92,457.31
Arkansas	\$77,637.36	New Hampshire	\$114,680.66
California	\$111,632.93	New Jersey	\$119,305.58
Colorado	\$107,936.13	New Mexico	\$71,531.93
Connecticut	\$117,303.22	New York	\$105,571.94
Delaware	\$102,639.68	North Carolina	\$84,727.74
Florida	\$85,581.31	North Dakota	\$90,647.13
Georgia	\$84,224.69	Ohio	\$90,396.68
Hawaii	\$105,978.87	Oklahoma	\$84,974.15
Idaho	\$91,721.86	Oregon	\$107,795.68
Illinois	\$103,958.50	Pennsylvania	\$99,681.52
Indiana	\$87,139.09	Rhode Island	\$98,980.03
Iowa	\$86,536.71	South Carolina	\$83,649.63
Kansas	\$96,719.61	South Dakota	\$83,574.96
Kentucky	\$87,474.11	Tennessee	\$81,911.63
Louisiana	\$78,124.94	Texas	\$98,362.04
Maine	\$84,312.62	Utah	\$112,799.70
Maryland	\$125,053.40	Vermont	\$95,683.34
Massachusetts	\$127,460.73	Virginia	\$114,127.44
Michigan	\$92,073.46	Washington	\$110,680.46
Minnesota	\$104,195.36	West Virginia	\$72,857.68
Mississippi	\$65,648.61	Wisconsin	\$90,695.26
Missouri	\$83,396.09	Wyoming	\$84,500.23

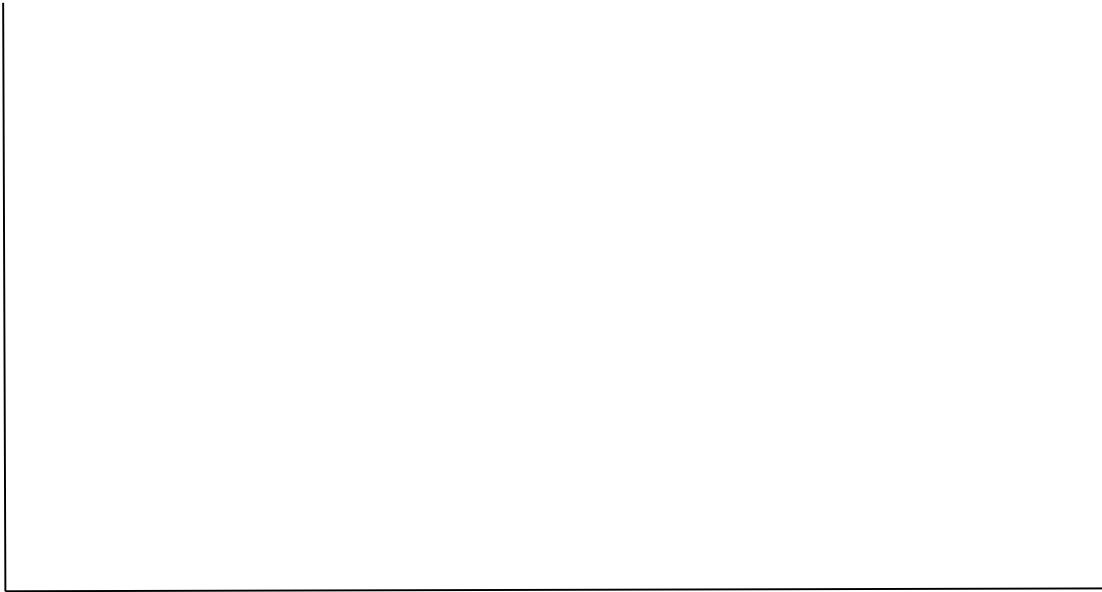
To make a histogram, we need to put the data into “bins” (even intervals that capture our data). We will do this first by hand by counting how many data scores are in each bin.

Lowest Income: _____ Highest Income: _____ Bin Width: _____

Interval	Count
\$65,000 - <\$77,600	
\$77,600 - <\$90,200	
\$90,200 - <\$102,800	
\$102,800 - <\$115,400	
\$115,400 - <\$128,000	

To make the histogram:

- Draw rectangles for each bin, and their height extends up to the highest count
- Bars MUST touch
- Label with each boundary with the "border value"



You can also use a graphing calculator to create a histogram, which we will do a little later!

Graphing Quantitative Variables: Stem and Leaf

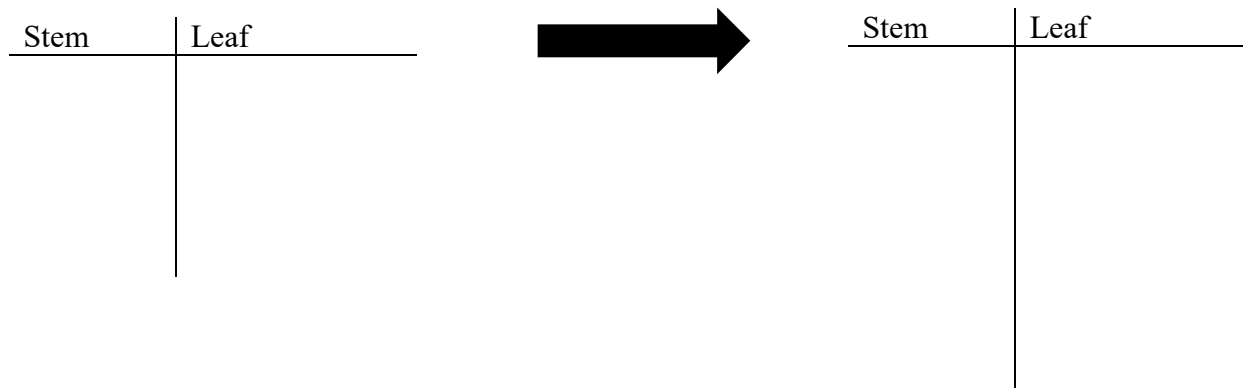
Gather Data: Determine your pulse rate. Hold the fingers of one hand on the artery on the inside of your wrist. Count the number of pulse beats in 20 seconds and then triple it. You will do this three times. Find the average of your tripled rate:

Pulse rate 1:	Pulse rate 1 tripled:
Pulse rate 2:	Pulse rate 2 tripled:
Pulse rate 3:	Pulse rate 3 tripled:
	Average:

Come to the board and write your average pulse along with your gender (M or F). When everyone has done so, write down all the class's data in the chart below.

Split Stem and Leaf

The last type of stem and leaf plot is called split stem and leaf. When you have too many data values in a single stem, it can be helpful to split the stem; the same way we would create more bins on a histogram if our bin width resulted in a skyscraper.



Here are the scores from last year on the Unit 1 Test:

80	88	78	93	69	80	92	90	88	84	82	92	88	62	84
75	74	96	88	88	90	94	92	79						

Create a split stem and leaf plot of the data.

Graphing Quantitative Variables: Dot plot

A **dotplot** is a very simple type of graph that involves plotting the data values, with dots, above the corresponding values on a number line.

How to construct a dotplot:

Step 1: Label your axis and title your graph. Draw a horizontal line and label it with the variable.

Step 2: Scale the axis based on the values of the variable.

Step 3: Mark a dot above the number on the horizontal axis corresponding to each data value.

Here are the quiz scores (out of 50) on a past quiz. Construct a dotplot.

36 42 42 42 42 41 38 41 40 42 43 44 45 45 46
 44 43 48 43 42 43 41 50 38 38 40 44

Graphing Quantitative Variables: Cumulative Relative Frequency Graphs

Cumulative relative frequency graphs (ogives) display percentiles.

A **percentile** will tell you what percent of data falls below a value.

You first must make a table of the cumulative relative frequencies in order to graph it. This can be done by finding the relative frequencies and then find the cumulative percentages.

Here is a frequency table summarizing the ages of the 46 U.S. presidents when they took office. (Biden was the 46th president and the oldest sworn in at 78 years old. Teddy Roosevelt, the 26th president, was the youngest at 42 years old.)

First Way: Find the relative frequencies and then find the cumulative percentages.

Age	40-<45	45-<50	50-<55	55-<60	60-<65	65-<70	70-<75	75-<80
Frequency	2	7	13	12	7	3	1	1
Relative Frequency								
Cumulative Relative Frequency								

Age	Cumu. Rel. Frequ
40-<45	
45-<50	
50-<55	
55-<60	
60-<65	
65-<70	
70-<75	
75-<80	

Creating an ogive:

- The first interval is 40-<45, with a cumulative relative frequency of 4%. Since 40 is the starting interval, that starts at 0 and 4% is graphed at 45.
- The next interval is 45-<50, with a cumulative relative frequency of 20%. At 45, 4% is graph, so at 50, the 20% will be graphed.
- This pattern continues until 100% is graphed at 80.

Questions we can answer with an ogive:

1) At approximately what age is the 30th percentile? What does this mean in the context of the problem?

2) Richard Nixon was the 37th president and was 56 years old when he was inaugurated. Approximately what percentile is this and what does it mean in the context of the problem?

